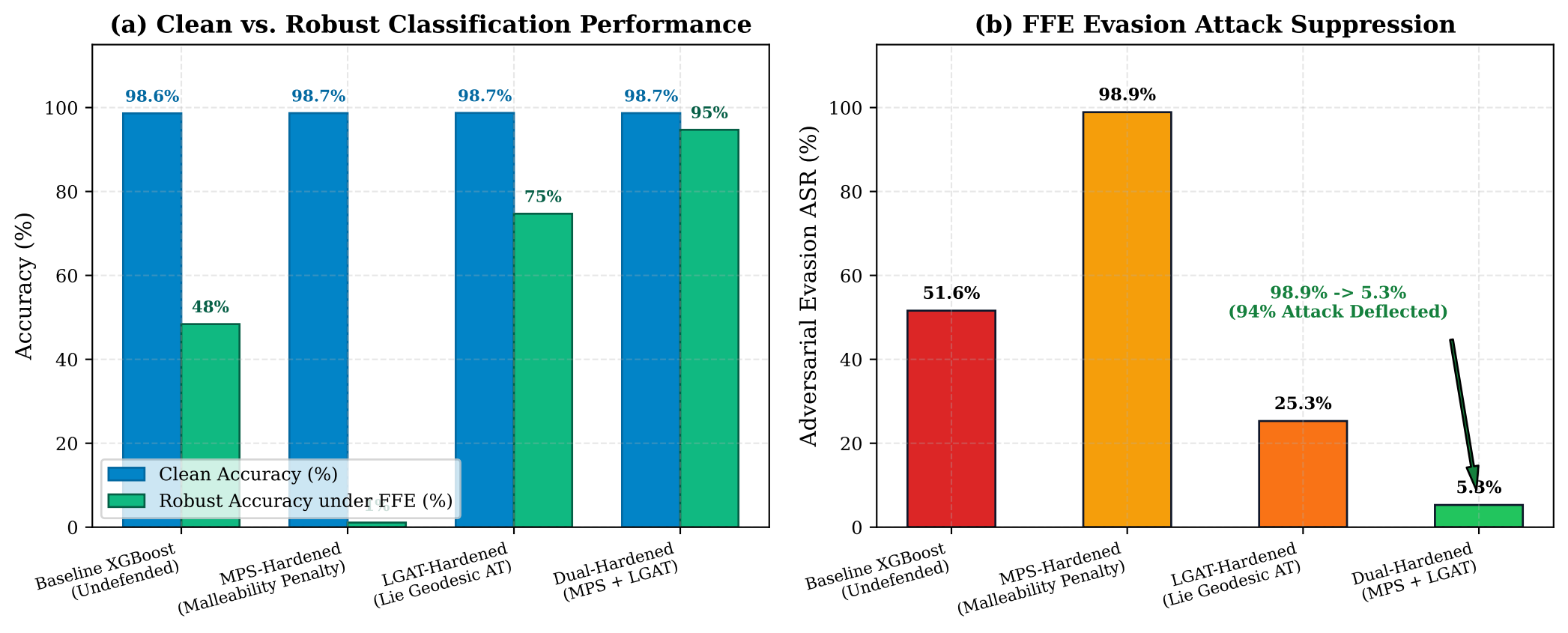
(a) Clean vs. Robust Classification Performance
Accuracy (%)
98.6%
48%
98.7%
98.7%
75%
98.7%
95%
Clean Accuracy (%)
Robust Accuracy under FFE (%)
Baseline XGBoost (Undefended)
MPS-Hardened (Malleability Penalty)
LGAT-Hardened (Lie Geodesic AT)
Dual-Hardened (MPS + LGAT)
(b) FFE Evasion Attack Suppression
Adversarial Evasion ASR (%)
51.6%
98.9%
98.9% -> 5.3%
(94% Attack Deflected)
25.3%
5.3%